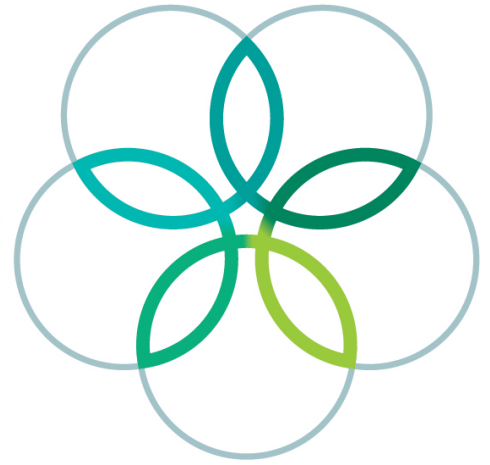


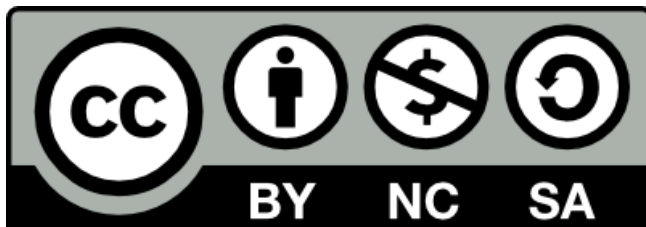
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25th IBO, Bali / Indonesia Practical Exams 1-4 ANSWERS

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25th INTERNATIONAL BIOLOGY OLYMPIAD

5 – 13 July, 2014

INDONESIA



PRACTICAL TEST 1 CELL & MOLECULAR BIOLOGY ANSWER KEY

Total points: **38.5**

Duration: 90 minutes

COUNTRY:	
STUDENT ID:	

Practical Test 1 (38.5 points)

Plasmid Identification and Telomere Analysis

Part A. Confirmation of plasmid sample X, Y, and Z by restriction analysis and DNA electrophoresis. (14 points)

Q 1.1. 8 points

For row 4 & 5 – 6 points; 1 series with two enzyme and 1 series *EcoRI*

For row 2 – 1 point if students fill 1 (μl) for every cell

For row 1 - 1 point for correct volume of water to add up to final volume of 10 μl

Table I. Design of Experiment for Plasmid Identification

No.	Reagents	Series 1 (Volume in μL)			Series 2 (Volume in μL)		
		Plasmid 1	Plasmid 2	Plasmid 3	Plasmid 1	Plasmid 2	Plasmid 3
		S1	S2	S3	S4	S5	S6
1	Sterile water	7	7	7	6	6	6
2	10 X Restriction buffer solution	1	1	1	1	1	1
3	DNA Plasmid	1	1	1	1	1	1
4	<i>EcoRI</i> *	1	1	1	1	1	1
5	<i>HindIII</i> *	0	0	0	1	1	1
	Volume total	10	10	10	10	10	10

* If the enzyme is used in the reaction, add 1 μL enzyme

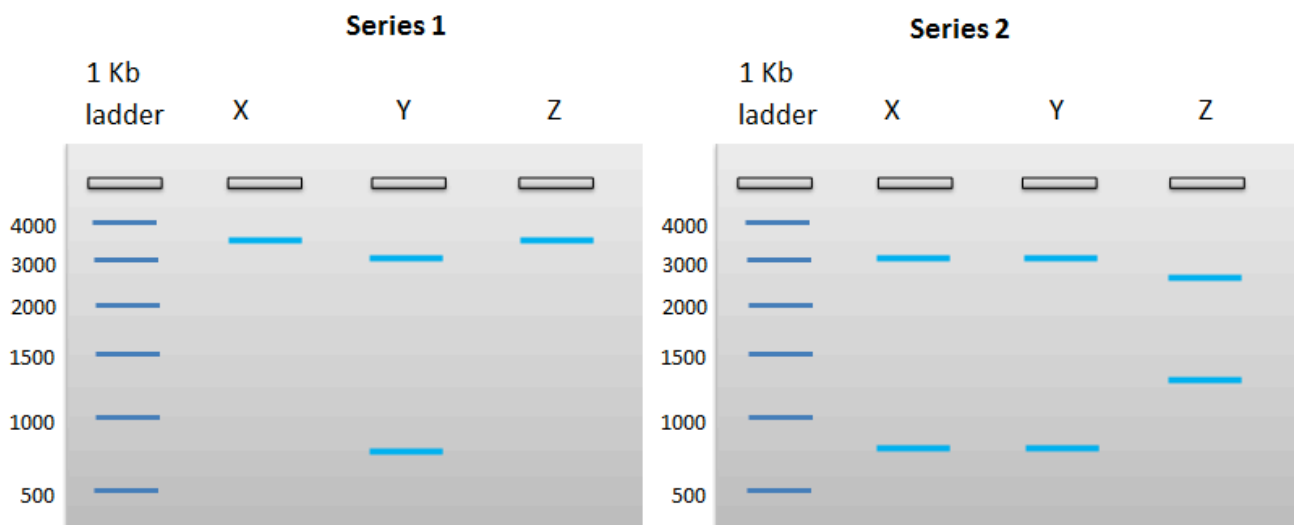
Or

No.	Reagents	Series 1 (Volume in μL)			Series 2 (Volume in μL)		
		Plasmid 1	Plasmid 2	Plasmid 3	Plasmid 1	Plasmid 2	Plasmid 3
		S1	S2	S3	S4	S5	S6
1	Sterile water	6	6	6	7	7	7
2	10 X Restriction buffer solution	1	1	1	1	1	1
3	DNA Plasmid	1	1	1	1	1	1

4	<i>EcoRI</i> *	1	1	1	1	1	1
5	<i>HindIII</i> *	1	1	1	0	0	0
	Volume total	10	10	10	10	10	10

Q1.2 Answers . (6 points – 1 points each lane)

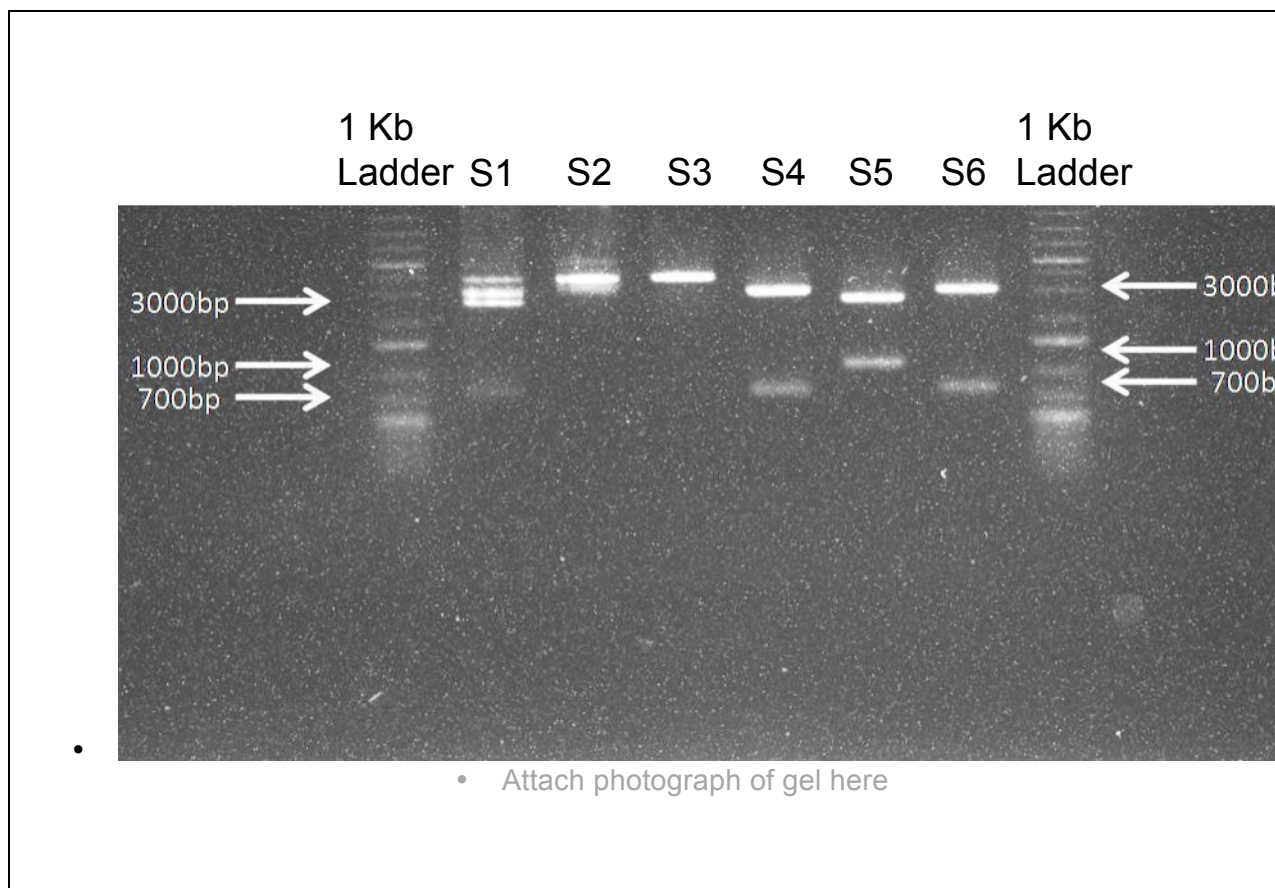
enzymes	Series 1	Series 2
<i>HindIII</i> A		✓
<i>EcoR1</i>	✓	✓



[Q 1.3 DNA Electrophoresis Result of Plasmid Restriction Experiment]

NOTE ABOUT THIS EXAM VERSION

Initially, the following plasmid restriction and electrophoresis had been intended as "question 1.3", with 26 points associated and the gel picture was intended as a central work output that contestants had to deliver. However, with regard to fairness and the way this exam section had to be enacted in IBO 2014, the 2014 IBO Jury decided not to count this practical part into the final ranking. This is why no points are indicated for this section of Practical Exam 1.



Code in problem sheet	Plasmid 1	Plasmid 2	Plasmid 3
Real size	3750	3786	3600
EcoRI restriction	3000 + 750	3786 (linearized)	3600 (linearized)
HindIII restriction	3750	3786 (linearized)	uncut
EcoRI + HindIII restriction	3000 + 750	2686 + 1100	2862 + 738

Part B. Cell reproduction and telomere analysis of *Paramecium* (24.5 points)

Q 2.1. (1 points x 2 = 2 points)

Culture	Binary Fission	Conjugation
A	√	-
B	√	√

Q 2.2. (2.5 points)

Day	Cell Concentration (Cell/mL)			Average cell concentration (cells/mL) N (Q2.2)	Log of Average cell concentration
	A	B	C		
0	1	1	1	1	0.00
1	8	10	16	11	1.05
2	80	120	128	109	2.04
3	640	960	1024	875	2.94
4	5760	7680	10240	7893	3.90

$$N = N_0 2^n$$

$$\log N = \log N_0 + n \log 2$$

$$\log N - \log N_0 = n \log 2$$

$$n = (\log N - \log N_0) / \log 2$$

n = fission rate (per day)

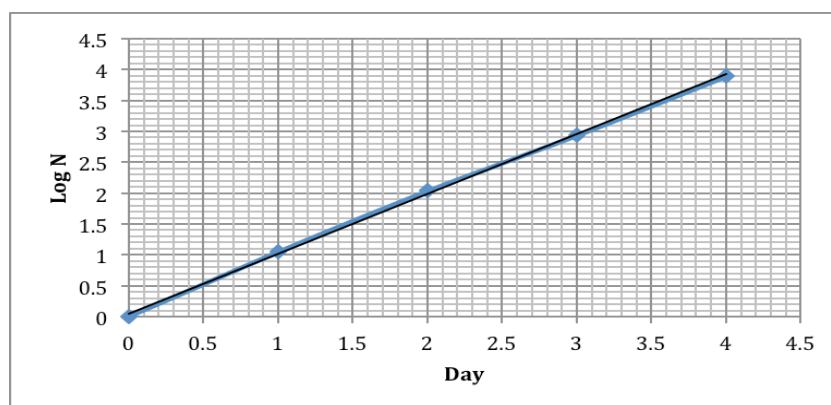
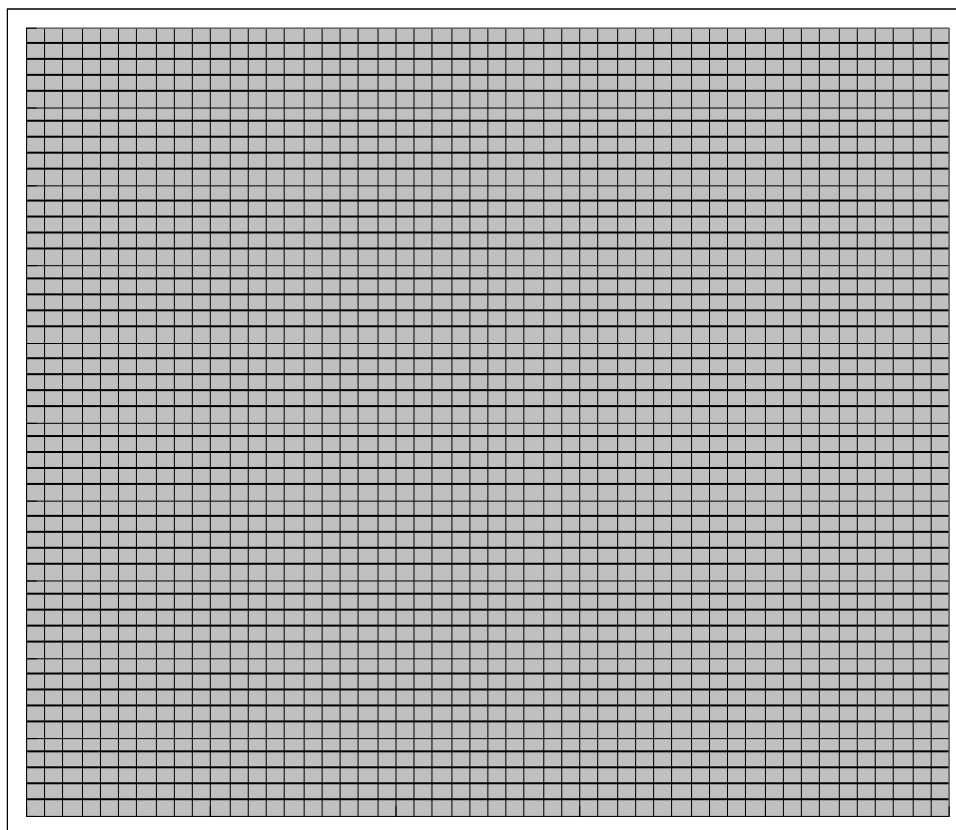
x = day;

N = average cell number

G= 1 day/fission rate →

Graph 8 points:

- **Axis X and Y : 2 points (1 point for each axis)**
- **5 points : 1 point for each value of log N**
- **1 point for drawing the line**



Q 2.3 (12 points)

No.	True	False
a	✓	
b		✓
c		✓
d	✓	
e		✓
f	✓	

25th INTERNATIONAL BIOLOGY OLYMPIAD

5 – 13 July, 2014

INDONESIA



PRACTICAL TEST 2 PLANT ANATOMY AND PHYSIOLOGY ANSWER KEY

Total points: **96**

Duration: 90 minutes

COUNTRY:	
STUDENT:	

The answers have to be given either with a tick (✓) or with Arabic numbers. The numbers "1" and "7" can look very similar in handwriting. To make sure that those two numbers can be well distinguished by the IBO staff, please write them as you normally would into the following box.

1 =		7 =	
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Task 1: Determination of plant pigment (36 points)

TLC plate photograph (4 points).

	4	3	2	1
- Did the experiment (1 point) and TLCed both samples (1 point) - Properly marked the scale on the TLC plate - Solvent reached the top end of the plate 4 pigment spots appear on the plate	all parameters are correct	Three parameters are correct	Two parameters are correct	Only one parameter is correct

Q 1.1 (12 points)

Spot	R _f values of four major pigments (precision: two places after the decimal point) (@ 2 point)	Pigment number from Table 1 (@ 1 point)
1	0.73 – 0.89 (0.81) → 2 0.66 (0.73 – 0.89) 0.98 → 1	β-carotene
2	0.15 – 0.17 (0.16) → 2 0.14 (0.15 – 0.17) 0.19 → 1	Xanthophyll
3	0.12 – 0.14 (0.13) → 2 0.11 (0.12 – 0.14) 0.15 → 1	Chlorophyll a
4	0.10 – 0.12 (0.11) → 2 0.09 (0.10 – 0.12) 0.13 → 1	Chlorophyll b

Q. 1.2 (4 points)

True	False
	✓
✓	
✓	
	✓

Q 1.3 (10 points)

Group 1

Extract	A		Total Chlorophyll (mg/L)	Chlorophyll a (mg/L)	Chlorophyll b (mg/L)
	649 (nm)	665 (nm)			
C	0.038 – 0.041	0.083 – 0.090	1.2663 - 1.369	0.91822 - 0.99684	0.3413 - 0.3648
D	0.111 – 0.117	0.273 – 0.281	3.8853 - 4.0541	3.10074 - 3.17578	0.7617 - 0.8549

Group 2

Extract	A		Total Chlorophyll (mg/L)	Chlorophyll a (mg/L)	Chlorophyll b (mg/L)
	649 (nm)	665 (nm)			
C	0.043 – 0.050	0.088 – 0.105	1.3968 - 1.6405	0.9579 -1.1505	0.4318 - 0.4815
D	0.100 – 0.111	0.245 – 0.263	2.6771 - 6.5043	2.7805 - 2.9637	0.6935 - 0.8387

Group 3

Extract	A		Total Chlorophyll (mg/L)	Chlorophyll a (mg/L)	Chlorophyll b (mg/L)
	649 (nm)	665 (nm)			
C	0.033 – 0.052	0.073 – 0.096	1.1053 - 1.6256	0.81002 - 1.01568	0.2893 - 0.6024

D	0.111 – 0.127	0.264 – 0.282	3.8304 - 4.2602	2.97744 - 3.13188	0.831 - 1.1052
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Group 4

Extract	A		Total Chlorophyll (mg/L)	Chlorophyll a (mg/L)	Chlorophyll b (mg/L)
	649 (nm)	665 (nm)			
C	0.029 – 0.043	0.068 – 0.082	0.9948 - 1.3602	0.76456 - 0.87572	0.2246 - 0.478
D	0.123 – 0.131	0.291 – 0.296	4.2351 - 4.4256	3.27822 - 3.30064	0.9327 - 1.1006

Q 1.4 (2 points)

Group 1

Extract	Ratio of chlorophyll
C	2.2184 - 2.3894
D	3.5337 - 4.0094

Group 2

Extract	Ratio of chlorophyll
C	2.6904 - 2.7326
D	3.7148 - 4.0708

Group 3

Extract	Ratio of chlorophyll
C	1.6860 - 2.7999
D	2.8338 - 3.5830

Group 4

Extract	Ratio of chlorophyll
C	1.8321 - 3.4041
D	2.9989 - 3.5148

Q 1.5 (4 points)

True	False
✓	
	✓
✓	
✓	

Task 2: Determination of starch content in root extract (21 points)

Q 2.1 (1 point)

Starch [ppm]	100
Starch solution (μL)	400
H ₂ O (μL)	600

Q 2.2 (8 points)

Group 1

Sample	Absorbance (580 nm)
Starch 250 ppm (C4)	1.198 – 1.208 → 2 1.078(1.198 – 1.208) 1.329 → 1
Starch 100 ppm (C6)	0.484 – 0.486 → 2 0.436(0.484 – 0.486) 0.535 → 1

Sample C7	0.375 – 0.401 → 2 0.338(0.375 – 0.401) 0.441 → 1
Sample C8	0.825 – 0.839 → 2 0.743(0.825 – 0.839) 0.923 → 2

Group 2

Sample	Absorbance (580 nm)
Starch 250 ppm (C4)	1.204 – 1.235 → 2 1.084(1.204 – 1.235) 1.358 → 1
Starch 100 ppm (C6)	0.485 – 0.494 → 2 0.437(0.485 – 0.494) 0.543 → 1
Sample C7	0.406 – 0.428 → 2 0.365(0.406 – 0.428) 0.471 → 1
Sample C8	0.843 – 0.863 → 2 0.759(0.843 – 0.863) 0.949 → 1

Group 3

Sample	Absorbance (580 nm)
Starch 250 ppm (C4)	1.136 – 1.154 → 2 1.022(1.136 – 1.154) 1.269 → 1
Starch 100 ppm (C6)	0.445 – 0.459 → 2 0.401(0.445 – 0.459) 0.505 → 1
Sample C7	0.362 – 0.377 → 2 0.326(0.362 – 0.377) 0.415 → 1
Sample C8	0.778 – 0.805 → 2 0.700(0.778 – 0.805) 0.886 → 1

Group 4

Sample	Absorbance (580 nm)
Starch 250 ppm (C4)	1.158 – 1.163 → 2 1.042(1.158 – 1.163) 1.279 → 1

Starch 100 ppm (C6)	0.457 – 0.466 → 2 0.411 (0.457 – 0.466) 0.513 → 1
Sample C7	0.382 – 0.396 → 2 0.344 (0.382 – 0.396) 0.436 → 1
Sample C8	0.795 – 0.812 → 2 0.716 (0.795 – 0.812) 0.893 → 1

Q 2.3 (4 points)

Q 2.3 (4 points)

a: $4.45 \times 10^{-3} - 0.01$

Q 2.4 (4 points)

Group 1 (02):

C7: 78 – 83 ;

C8: 171 – 174

Group 2 (01):

C7: 84 – 87 ;

C8: 174 – 175

Group 3 (03):

C7: 79 – 82 ;

C8: 171 – 175

Group 4 (04):

C7: 82 - 85 ;

C8: 171 – 175

Q 2.5 (4 points)

True	False
√	
	√
√	
√	

Task 3. Observation of structural adaptation in plants (39 Points)

Q 3.1 (6 points @ 0.5 point)

Tissue type	Presence		
Specimen	X	Y	Z
Cortex			
a. sclerenchyma	+	+	-
Endodermis	+	+	+
Xylem			
a. primary xylem	+	+	+
b. secondary xylem	-	-	-

Q. 3.2 (9 points)

Specimen	Diagram (number)
X	4
Y	2
Z	6

Q 3.3 (9 points)

Specimen	No aerenchyma	Lysigenous*	Schizogenous**
X	√		
Y		√	
Z			√

* This type of intercellular space arises through dissolution of entire cells.

** This type of intercellular space arises through separation of cell walls from each other along more or less extended areas of their contact.

Q 3.4 (9 points)

Organ	Monocotyledonous			Dicotyledonous		
	Root	Stem	Leaf	Root	Stem	Leaf
X	✓					
Y	✓					
Z				✓		

Q 3.5 (6 points)

Specimen	Control	Flooding
X	✓	
Y		✓
Z		✓

End of the Practical Exam

25th INTERNATIONAL BIOLOGY OLYMPIAD

5th – 13th July, 2014

INDONESIA



PRACTICAL TEST 3

ANIMAL ANATOMY, PHYSIOLOGY AND SYSTEMATICS

ANSWER SHEET

Total points: **93.5**

Duration: 90 minutes

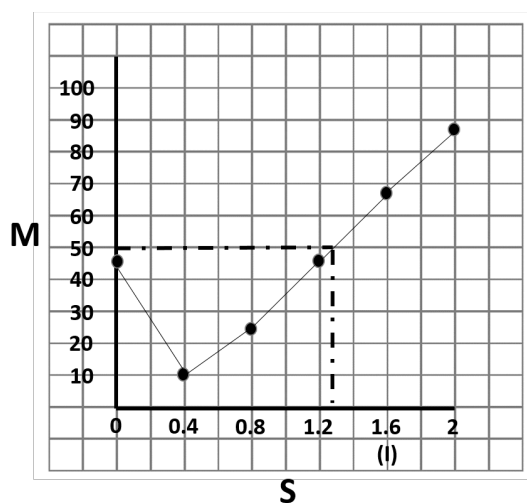
COUNTRY:	
STUDENT:	

Task (93.5 points)

Acute Response of Fish Larvae to Rapid Changes in Salinity to Salt Concentration, Calculation of LC_{50} , and Classification of Prawns

Part A. Acute Response of Fish Larvae to Rapid Changes in Salinity (12 points)

Q 1.1 and Q.1.2 (6 + 2 points)



Q 1.3. (2 points)

Mark the appropriate answer with a tick "✓"

A	B	C	D
			✓

Q 1.4. (2 points)

Mark the appropriate answer with a tick "✓"

A	B	C	D
			✓

25th INTERNATIONAL BIOLOGY OLYMPIAD

5 – 13 July 2014

INDONESIA



PRACTICAL TEST 4

ECOLOGY AND ETHOLOGY

ANSWER KEY (Rev. 09-07-14)

Total points: **100**

Duration: 90 minutes

COUNTRY:	
STUDENT:	

The answers have to be given either with a tick (✓) or with Arabic numbers. The numbers "1" and "7" can look very similar in handwriting. To make sure that those two numbers can be well distinguished by the IBO staff, please write them as you normally would into the following box.

1 =		7 =	
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Biogeography and Biodiversity

TASK 1. ISLAND BIOGEOGRAPHY

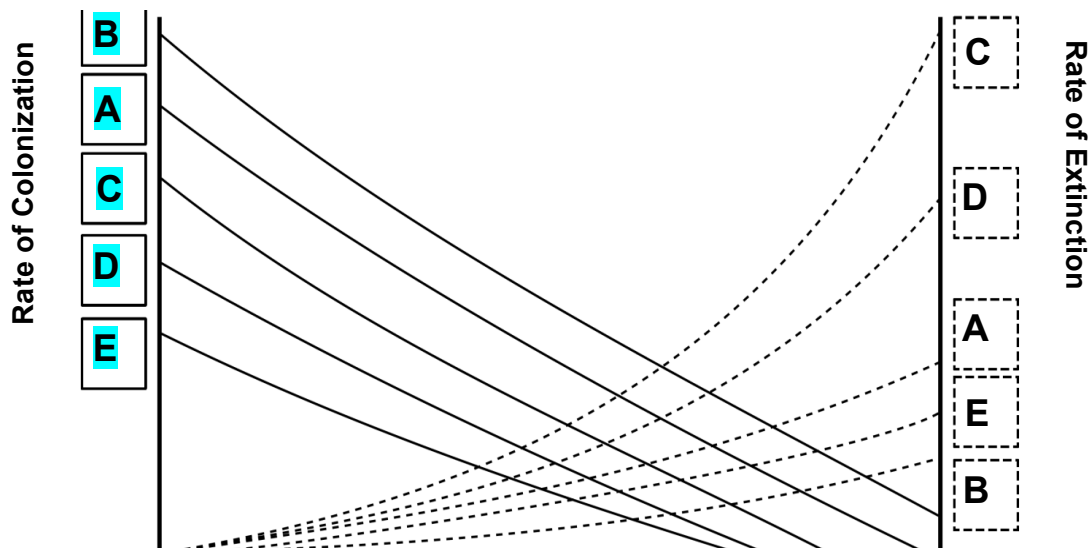
Question 1.1 (5 points: 5 x 1) (straightforward – no partial points) values slightly outside the range were still accepted.

No.	Island	Distance from mainland (km) (accepted range in parentheses)
1.	Ternate	12.8 (12.4 – 13.2)
2.	Tidore	9.7 (9.3 – 10.1)
3.	Mare	13.0 (12.6 – 13.4)
4.	Moti	14.1 (13.7 – 14.5)
5.	Makian	17.8 (17.4 – 18.2)

Grading Rule:

- Measurements within range are accepted for 1 point each, no partial points.
- *Range determined as mean of 20 measurements that have been made during our simulation, +/- standard deviation which reached 0.35. Therefore, the accepted range was determined as mean +/- 0.40.

Question 1.2 (10 points: 10 x 1) (straightforward)



Number of Species in Equilibrium

Question 1.3 (2 points: 2 x 1) (straightforward)

Place a tick (✓) mark in the box of your selected correct answer.

A	B	C	D	E
✓	✓			

TASK 2. PRIMARY SUCCESSION AFTER VOLCANIC ERUPTION

Part A : Succession and Plant Community Structure

Question 2.1 (3 points) (straightforward–no partial points)

Place a tick (✓) mark in the box of your selected correct answer

A	B	C	D	E
			✓	

Question 2.2 (5 points; 10 x 0.5) (straightforward–no partial points)

Fill in your calculated similarity indices in the boxes provided.

Year	1934	1949	1963	1979	1991
1949	0.60				
1963	0.52	0.32			
1979	0.48	0.29	0.94		
1991	0.52	0.26	0.89	0.95	

Question 2.3 (2 points) (straightforward–no partial points)

Place a tick (✓) mark in the box of your selected correct answer.

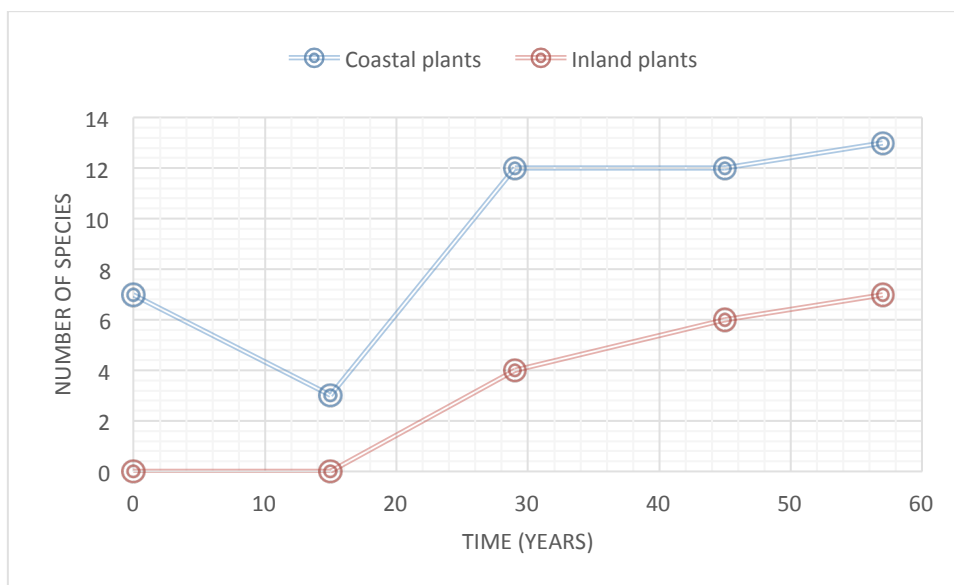
A	B	C	D	E
				✓

Question 2.4 (4 points: 4 x 1)

Place a tick (✓) mark in the box of your selected correct answer. (straightforward–no partial points)

Statement	True	False
A		✓
B		✓
C	✓	
D		✓

Question 2.5 (6 points)



Grading Rule

- Correct format of graph using correct data from table (two separate graphs for coastal and inland plants will also be accepted) (5 points)
- correct axes and labels (1 point)

Question 2.6 (2 points: 2 x 1) (straightforward–no partial points)

Place a tick (✓) mark in the box of your selected correct answer.

Statement	True	False
A		✓
B	✓	

Part B : Dispersal Biology of *Ficus*

Question 2.7. (15 points) Answer data may vary slightly among participants)

Fruit Serial Number	Fruit Diameter (mm)		
	<i>Ficus hispida</i>	<i>Ficus septica</i>	<i>Ficus variegata</i>
1	32.12		
2			22.54
3		21.22	
4			20.42
5		20.56	
6		18.74	
7			21.45
8		19.63	
9	28.34		
10			24.21
11		23.22	
12	30.46		
13		21.43	
14			21.55
15	29.12		
16		24.24	
17	27.50		
18		20.45	
19			24.54
20			22.11
21		18.12	
22			
23	25.12		
24			23.50
25	26.42		
26			20.22
27		22.86	
28	31.22		
29			23.42
30		17.15	
Mean	28.79	21.05	22.28
Std. Dev.	2.42	2.21	1.51

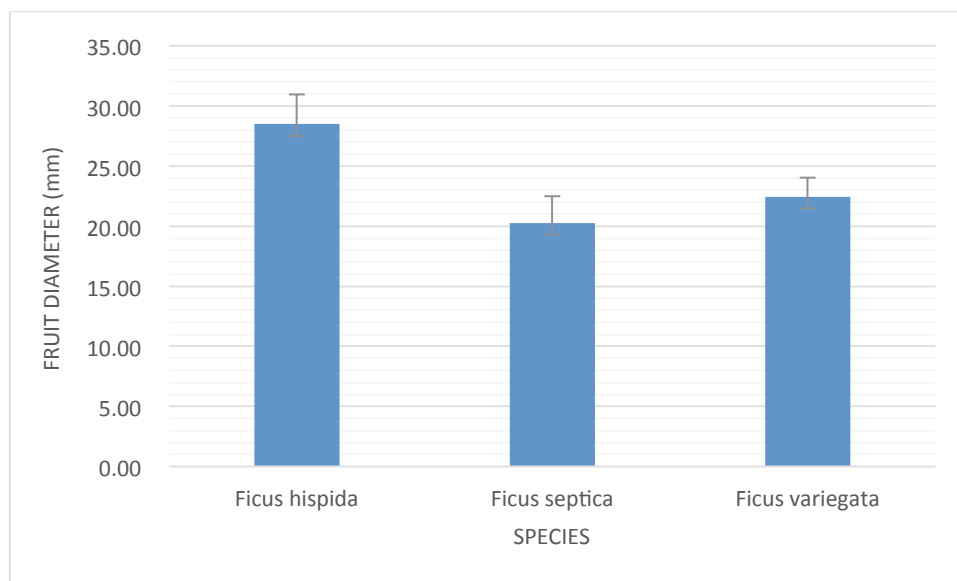
*Measurements should be with two decimal places.

Note: Standard deviation (square root of variance) using either formula (divided by “n” as in the variance formula given, or divided by “n-1”) were accepted.

Grading Rule

- Each measurement = 0.25 point (total 7.5 points).
- Mean and standard deviation = 2.5 points for each species.
- Measurements will be compared to our own record of measurements.

Question 2.8. (8 points) (Example)



Grading Rule:

- Correct format of bar graph (4 points)
- Graph based on correct values from Question 2.7 (2 points)
- Standard deviation given correctly (1 point)
- Vertical axis given correct label and units (1 point)

Question 2.9 (4 points: 4 x 1)

Place a tick (✓) mark in the box of your selected correct answer: (straightforward – no partial points)

Statement	True	False
A	✓	
B	✓	
C	✓	
D		✓

Question 2.10 (3 points: 3 x 1) (straightforward – no partial points)

Place a tick (✓) mark in the box of your selected correct answer:

Statement	True	False
A	✓	
B		✓
C	✓	

TASK 3. SPECIATION IN SONGBIRDS

Question 3.1 (1 point: 2 x 0.5) (straightforward – no partial points)

Place a tick (✓) mark in the box of your selected correct answer.

Note: Students were given full points if they tick the two correct answers, and were not penalized if they additionally ticked one other answer which is incorrect.

A	B	C	D	E
		✓		✓

Question 3.2 (18 points: 9 x 2) (straightforward–no partial points)

Songbird Number	Syllable Repertoire	Acceptable Range for Answers
1	11	9 – 13
2	19	17 – 21
3	30	28 – 32
4	20	18 – 22
5	29	27 – 31
6	11	9 – 13
7	32	30 – 34
8	12	10 – 14
9	24	22 – 26

Question 3.3 (3 points: 3 x 1)

Group	Songbird Number
...	1, 6, 8
...	2, 4, 9
...	3, 5, 7

Grading Rule:

- Grading within a group will only be based on the first three numbers (in the case where students enter more than three songbird numbers in a group).
- Full point (1) given if all three numbers are correct.
- Partial point (0.5) given if two numbers are correct.
- No points (0) given if only one number or none is correct.

Question 3.4 (8 points: 2 x 4)

Note: statistical calculations were graded very leniently. Two formulas for variance (divided by “n” as in the formula given, or divided by “n-1”) were accepted.

Since it was not specified in the question, both one-tailed and two-tailed tests were accepted as correct.

t-values may vary according to the acceptable range for answers (number of different syllables).

	Group	Group
	11	30
	11	29
	12	32
Mean within group	11.33	30.33
Variance within group	0.11 0.22	0.78 1.56

Calculated t-value	20.152
Table t-value	2.132 (one-tailed) or 2.776 (for two-tailed test)

Grading Rule:

- 1 point each for mean and variance for two groups.
- 2 points each for calculated t-value and table t-value.

Question 3.5 (1 point) (straightforward – no partial points)

Place a tick (✓) mark in the box of your selected correct answer.

A	B
	✓

_____ END OF PAPER _____